

Prdm1 Positively Regulates Liver Type 1 ILCs Cancer Immune Surveillance and Preserves Functional Heterogeneity

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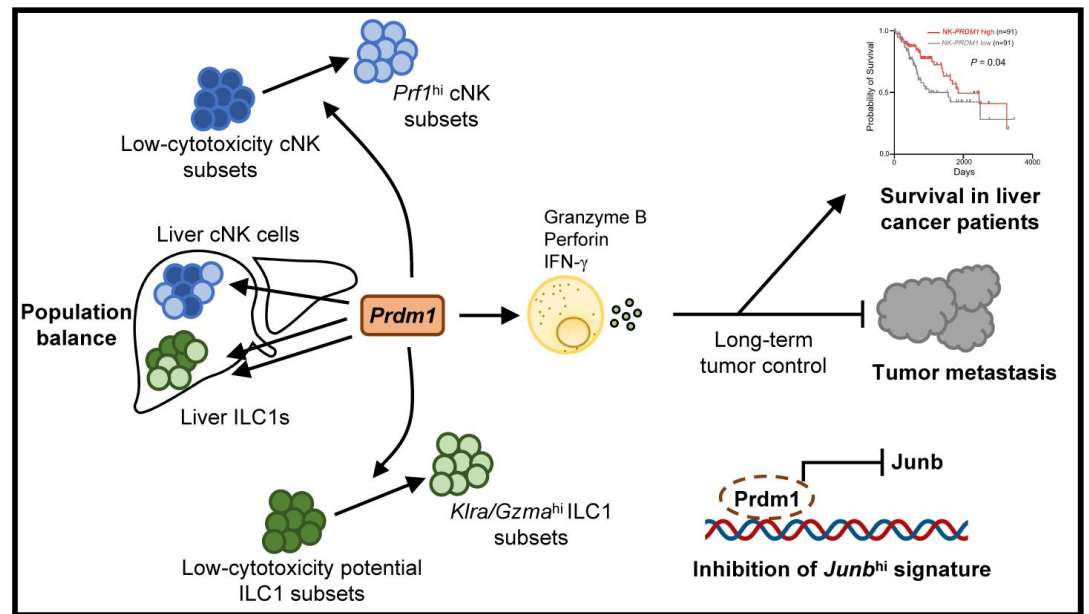
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Abstract

Liver Type 1 innate lymphoid cells (ILCs) comprise conventional natural killer (cNK) cells and ILC1s. The main functions of Type 1 ILCs not only include directly killing target cells but also regulating the local immune microenvironment of the liver through the secretion of cytokines. Uncovering the intricate mechanisms by which transcriptional factors regulate and influence the functions of Type 1 ILCs, particularly within the context of liver tumors, presents a significant opportunity to amplify the effectiveness of immunotherapies against liver malignancies. Using Ncr1-driven conditional knockout mouse model, our study reveals the regulatory role of *Prdm1* in shaping the composition and maturation of liver Type 1 ILCs. Notably, *Prdm1* regulates the ratio between NK cells and ILC1s, promoting a shift in the balance towards the direction of NK cells. Although *Prdm1* did not affect the killing function of cNK cells in an *in vivo* cytotoxicity model, a significant increase in cancer metastasis was observed in *Prdm1* knockout mice. IFN- γ , granzyme B, and perforin secretion decreased significantly in *Prdm1* deficient Type 1 ILCs. scRNA sequencing data also provided evidence that *Prdm1* sustains functional subsets of liver type 1 ILCs and facilitates communications between Type 1 ILCs and macrophages. The present study unveiled a novel regulatory mechanism of *Prdm1* in liver Type 1 ILCs, showing promising potential for developing innovative immune therapy strategies against liver cancer.

Graphical abstract



eLife assessment

The authors investigated the requirement and function of Blimp1/Prdm1 in murine natural killer (NK) cells and the ILC1 lineage of innate lymphoid cells, using a conditional knockout model. The single-cell mRNA-seq data provided here represent a **valuable** resource for the community, but the lack of mechanistic investigations leaves the study **incomplete**. The work will be of interest to the fields of innate lymphoid cell biology and tissue immunology.

Introduction

Liver Type 1 ILCs consist of cNK cells and ILC1s (1), with distinct developmental trajectories and effect molecules (2). Both NK cells and ILC1s play indispensable roles in combatting viral infections (3), maintaining local immune homeostasis (4), eradicating malignant transformed cells (5), and fostering cross-talk with adaptive immunity (6). NK cells demonstrate potent cellular cytotoxicity, facilitating direct elimination of target cells (7). On the contrary, a defining attribute of ILC1s is their predominant cytokine-mediated functions, with limited cellular killing capacity (8). In a state of homeostasis, liver Type 1 ILCs (CD45⁺CD3⁺NK1.1⁺NKp46⁺) can be discriminated into cNK cells and ILC1s by the differential expression of CD49a and CD49b (1): liver cNK cells are marked by the expression of CD49b, while liver ILC1s exhibit a distinctive positivity for CD49a. Tumor Necrosis Factor Related Apoptosis Inducing Ligand (TRAIL) is also expressed on liver ILC1s, but not on liver NK cells (9, 10). Transcriptional factors (TFs) such as T-bet, Nfil3, PLZF, and ID2 are also required for Type 1 ILCs development or generation of their progenitors (11–14). Eomes is considered to be necessary for NK cell maturation, but not for ILC1s (15). On the contrary, deficiency of Hobit results in the depletion of ILC1s, while only leaving little impact on liver NK cells (16). These studies suggest that TFs may, or usually, play different roles in NK cells and ILC1s.

During the development of liver cancers, the functions of immune cells are often inhibited, resulting in the formation of an immunosuppressive tumor microenvironment, and sometimes even systemic immune suppression (17 [↗](#)). Transforming a cold tumor into a hot tumor with a stronger immune response is one of the goals of many cancer immunotherapies (18 [↗](#)). Research targeting the immune system is showing increasing clinical promise in liver cancer treatment (19 [↗](#)). Our recent study also found that Toll-like receptor agonists can significantly enhance the anti-tumor effect of Sorafenib by reconstructing the tumor immune microenvironment and reshaping the vascular system (20 [↗](#)).

The anti-tumor activity of NK cells in the liver is relatively well-established, whereas the relationship between ILC1 and tumors is still a topic of controversy. Clinical research findings have shown that liver tumors with a higher infiltration of NK cells are associated with a better prognosis (21 [↗](#)). Decreased NK cell activity is closely correlated with the malignancy of liver tumors and represents a significant risk factor for recurrence (22 [↗](#)). The tumor microenvironment orchestrates the transformation of NK cells into ILC1s in a TGF- β -dependent manner, resulting in their diminished capacity to control tumor growth and metastasis. This process ultimately promotes tumor immunoevasion (23 [↗](#)). However, research has also revealed that, distinct from directly inhibiting tumor growth, the primary function of ILC1 is to suppress the seeding of metastatic tumor cells in liver tissue (5 [↗](#)). Comprehensive research in this field is essential to harness the precision of Type 1 ILCs targeted immunotherapy for liver cancers.

The transcription factor network governs the function of Type 1 ILCs and the balance between NK and ILC1. Our research team, along with others, has observed that Smad4 promotes the shift in balance from NK cells towards ILC1 in a TGF- β -independent pathway. This study also revealed that Smad4 positively regulates the expression of another transcription factor, Prdm1 (24 [↗](#), 25 [↗](#)). PR domain 1 (Prdm1/Blimp1) plays a crucial role in the differentiation of B cells into plasma cells and the homeostasis of T cells (26 [↗](#)–28 [↗](#)). The function and regulatory network of Prdm1 in NK cells is distinct from B cells and T cells, for its expression is independent of B-cell lymphoma 6 (Bcl-6) and Interferon Regulatory factor 4 (IRF4) but relies on T-bet. In the study that identified Prdm1 as an essential transcriptional factor for NK cell maturation (29 [↗](#)), no significant differences were observed in IFN- γ production or cytotoxicity between Prdm1-deficient and wild-type NK cells. Although Prdm1 expression is dependent on IL-15 in immature NK cells and can be further upregulated by IL-12 or IL-21, it plays a role in downregulating the expression of certain cytokine receptors, such as CD25, consequently diminishing the responsiveness of NK cells to IL-2 (30 [↗](#)). Notably, some studies have even suggested that Prdm1 suppresses the secretion of IFN- γ by NK cells (31 [↗](#)). These findings imply that, although Prdm1 promotes NK cell development, it may function as a negative regulator of their activity. However, direct evidence supporting the impact of Prdm1 on NK cell anti-tumor capabilities is still lacking. Furthermore, there has been no investigation into its influence on the homeostasis of NK cells and ILC1s in the liver, nor has there been an exploration of the underlying mechanisms.

In the current study, we found that the deletion of *Prdm1* in Ncr1⁺ cells resulted in an imbalance in the homeostasis of liver Type 1 ILCs, with a shift towards cNK cells. The data also support the essential role of Prdm1 in the cancer surveillance mediated by NK cells. While Prdm1 positively regulates genes associated with cellular cytotoxicity, it concurrently exerts inhibitory control over certain positive regulators of NK cell development and functionality, such as JunB. Using single-cell RNA sequencing, we have identified a subset of NK cells within the liver characterized by elevated JunB expression. These cells exhibit decreased expression of genes associated with cellular cytotoxicity, and their abundance significantly increases following Prdm1 knockout.

Results

Prdm1 promotes Type 1 ILCs homeostasis and terminal maturation

Examination of 363 liver hepatocellular carcinoma (LIHC) patient samples from The Cancer Genome Atlas (TCGA) revealed a positive correlation between the expression of NK cell-associated genes (*NCR1*, *NCR3*, *KLRB1*, *CD160*, and *PRF1*) (32) and *PRDM1* expression (Figure 1A). Patients with top and bottom quartiles of *NK-PRDM1* signature expression were chosen for survival analysis (Figure 1B). Notably, patients with the *NK-PRDM1*^{hi} signature had better overall survival compared to the these with *NK-PRDM1*^{lo} signature (Figure 1C). Similar results were also found in skin cutaneous melanoma (SKCM, n=454) and lung adenocarcinoma (LUAD, n=497) patients (Supplemental Figure 1, A-F). These data suggested that *PRDM1* in NK cells might be essential for immune surveillance in some solid tumors, including liver cancer. These findings prompted us to investigate the impact and mechanism of *PRDM1* in NK cells and ILC1 within the context of liver cancer.

To achieve this, we developed a conditional knockout mouse model to specifically deplete *Prdm1* in Type 1 ILCs. *Ncr1-Cre* mice were crossed with *Prdm1*^{fl/fl} mice to generate *Ncr1-cre Prdm1*^{fl/fl} mice, which specifically knockout exons 6-8 of *Prdm1* (33) in NKp46 positive cells (Figure 1D). The mice carrying *Ncr1*^{Cre/+}*Prdm1*^{fl/fl} were referred to as *Prdm1*^{ΔNcr1} mice, and the mice carrying *Ncr1*^{+/+}*Prdm1*^{fl/fl} were referred to as *Prdm1*^{+/+} mice or wild-type (WT) mice. To further validate the deletion of *Prdm1* in NKp46⁺ cells of *Prdm1*^{ΔNcr1} mice, CD3⁺NK1.1⁺NKp46⁺ cells were sorted and the expression of *Prdm1* was quantified by real-time RT-PCR. The expression of *Prdm1* was almost undetectable in *Prdm1*^{ΔNcr1} mice (Figure 1E), indicating successful knockout of *Prdm1* in NKp46⁺ cells.

Proportion and absolute number of cNK cells in blood, bone marrow, lung, liver, spleen, and lymph nodes were analyzed by flow cytometry. Compared with *Prdm1*^{+/+} mice, the percentage of cNK cells (CD3⁺NK1.1⁺NKp46⁺) among lymphocytes was decreased in all of these tissues except bone marrow and lymph nodes (Figure 1F; Supplemental Figure 2A). However, no significant difference was observed in the percentage of cNK cells among bone marrow-derived lymphocytes between *Prdm1*^{ΔNcr1} and *Prdm1*^{+/+} mice. The absolute number of cNK cells in blood, lung, liver, and spleen also decreased in *Prdm1*^{ΔNcr1} mice (Figure 1F; Supplemental Figure 2A). Only a slight decrease in the number of cNK cells was observed in the lymph nodes of *Prdm1*^{ΔNcr1} mice, which did not reach statistical significance either (Supplemental Figure 2A). In contrast, the absolute number of cNK cells in *Prdm1*^{fl/fl} mice bone marrow is moderately higher than *Prdm1*^{ΔNcr1} mice (Figure 1F).

NK cell terminal maturation can be divided into four stages according to the expression of CD11b and CD27 (34). Stage IV (CD11b⁺CD27⁺) NK cells show higher level of effector molecules than any other stages and are considered as the most mature NK cells (35). Based on the expression of CD11b and CD27, the maturation of cNK cells from blood, bone marrow, lung, liver, spleen, and lymph nodes were assessed. Compared with *Prdm1*^{+/+} mice, the proportion of most mature CD11b⁺CD27⁺ NK cells were significantly decreased in all of the analyzed tissues in *Prdm1*^{ΔNcr1} mice, without any exceptions (Figure 1G; Supplemental Figure 2B). Remarkably, an increase in immature CD11b⁺CD27⁺ NK cells was also observed in these tissues. Killer cell lectin-like receptor subfamily G member 1 (KLRG1) is a lectin-like receptor which was considered as another marker of NK cell maturation (36). In both *Prdm1*^{+/+} and *Prdm1*^{ΔNcr1} mice, cNK cells from the liver and lung had the highest expression of KLRG1, following by blood and spleen (Supplemental Figure 2, C and D). The lowest KLRG1 expression was observed in cNK cells derived from lymph nodes and bone marrow, indicating the presence of the most immature cNK cells in these tissues. Consistent with CD11b/CD27 based maturation analysis, a significant loss of KLRG1⁺

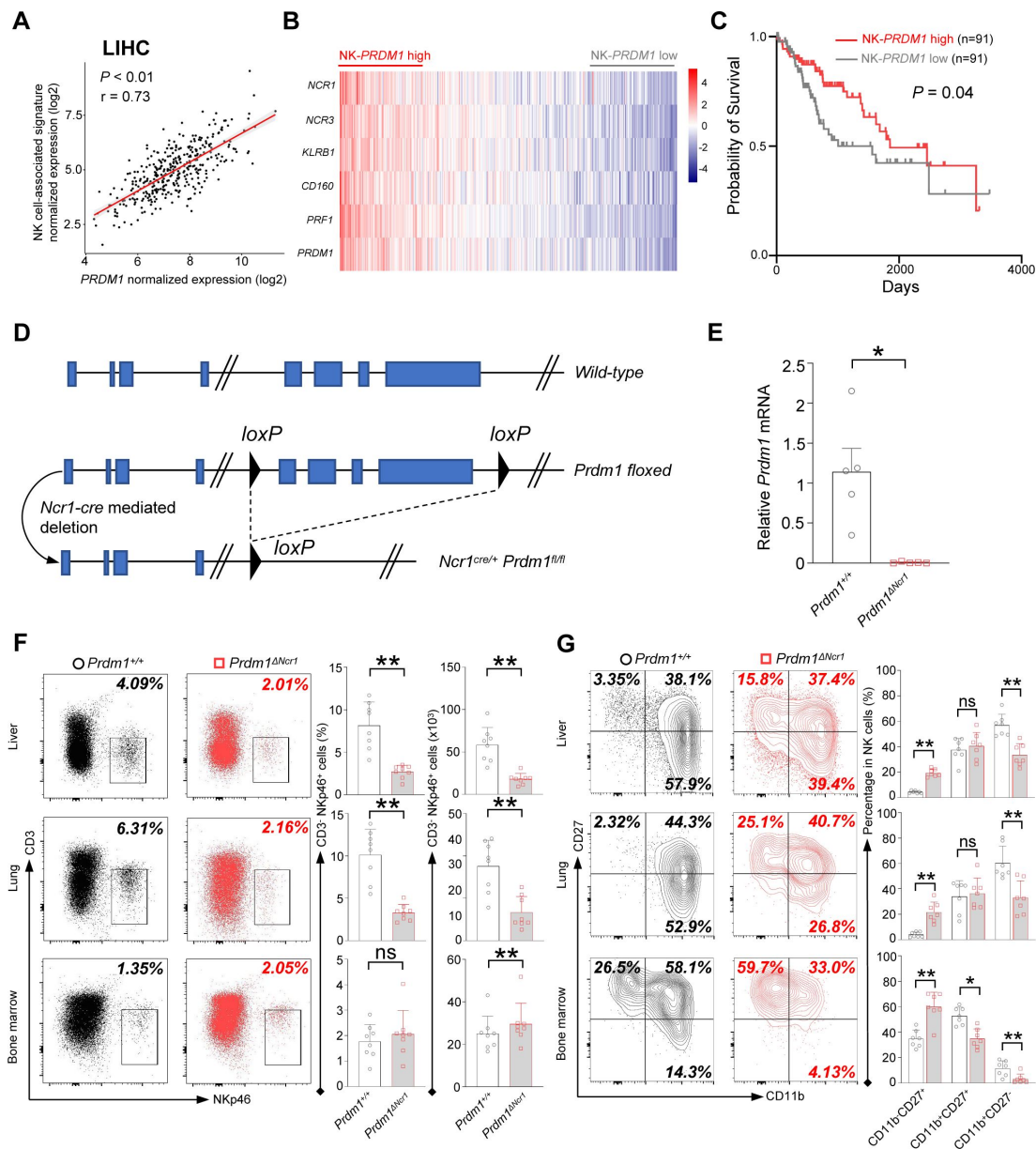


Figure 1.

***Prdm1* promotes type 1 ILCs homeostasis and terminal maturation.**

(A) Correlation between the average expression of NK cell-associated genes (*NCR1*, *NCR3*, *PRF1*, *CD160*, *KLRB1*) and *PRDM1* in LIHC (Liver Hepatocellular Carcinoma; n=363) patients sourced from TCGA datasets. (B) Heatmap of the ordered, z-score normalized expression values for *PRDM1* and NK cell-associated genes in liver cancer patients. The top and bottom quartiles of all patient samples are indicated. (C) Prognostic value of the NK-*PRDM1* signature for overall survival of liver cancer patients comparing high and low quartiles. (D) Schematic representation of the *Prdm1*-conditional knockout mouse model. Targeted exon 6-8 of *Prdm1* (top) is flanked with *loxP* sites (middle). *Ncr1*-expressed Cre recombinase was used to generate the *Prdm1*^{ΔNcr1} allele (bottom). (E) Real-time RT-PCR quantification of *Prdm1* expression in NKp46⁺ cells to determine the presence of *Prdm1* (n=5). (F) Representative flow cytometric plots (left) and quantification (right) of the proportion and absolute number of CD3⁺NKp46⁺ cells among lymphocytes in liver, lung, and bone marrow (n=8). (G) Representative flow cytometric plots (left) of the CD11b and CD27 expression within CD3⁺NK1.1⁺NKp46⁺ cells in liver, lung, and bone marrow (n=7). Right panel showed the percentage of distinct stages of NK cells. Data are presented as the mean±SEM and were analyzed by 2-tailed, paired t-test. Differences were evaluated between littermates. Each circle and square on graphs represents an individual mouse; *P*, *P*-value; *r*, pearson correlation coefficient; *, *P*<0.05; **, *P*<0.01, ns, not significant.

NK cells in *Prdm1*^{ΔNcr1} mice was observed compared to *Prdm1*^{+/+} mice (Supplemental Figure 2, C and D). Together, these data showed that *Prdm1* is required for the terminal maturation of cNK cells among various tissues.

***Prdm1* is required for Type 1 ILCs to control tumor metastasis**

Two subpopulations of liver Type 1 ILCs (gated by CD45⁺CD3⁺NK1.1⁺NKp46⁺) were further analyzed based on the expression of CD49a and CD49b (Figure 2A). Compared with *Prdm1*^{+/+} mice, the *Prdm1*^{ΔNcr1} mice exhibited an increased percentage of cNK cells (CD49a⁺CD49b⁺) and reduced proportion of ILC1s (CD49a⁺CD49b⁻) (Figure 2A). Of note, the absolute number of both cNK cells and ILC1s were decreased in *Prdm1*^{ΔNcr1} mice, with a more robustly reduction in ILC1s (Figure 2A). This underscored the crucial role of *Prdm1* in maintaining the homeostasis of both liver cNK cells and ILC1s.

The disruption of NK cell homeostasis and maturation due to *Prdm1* loss motivated us to further explore whether deficiency of *Prdm1* impaired NK cell cytotoxicity. A B2M-deficient cell-based *in vivo* cytotoxicity assay was used to evaluate the effect of *Prdm1* on the cytotoxicity of NK cells (37, 38). B2M-deficient cells do not have detectable Major Histocompatibility Complex I (MHC-I) on the cell surface, making them the target of NK cells (39). Healthy NK cells will reject B2M-deficient donor cells efficiently and the elimination was used to quantify the cytotoxicity of NK cells. Although significant impaired homeostasis and maturation of NK cells were observed in *Prdm1*^{ΔNcr1} mice, no significant difference in the *in vivo* cytotoxicity assay were observed between *Prdm1*^{+/+} and *Prdm1*^{ΔNcr1} mice (Figure 2, B and C).

While our B2M-deficient cell-based assay did not reveal a significant impact of *Prdm1* on NK cell cytotoxicity, both our results and those of other researchers have demonstrated *Prdm1*'s role in positively regulating NK cell development and homeostasis. Besides direct cytotoxicity, other factors influence the anti-tumor capability of NK cells, such as the ability to counteract tumor-induced immune suppression and exhaustion, the secretion of cytokines to activate other immune cells for tumor elimination. Moreover, the cytotoxicity assay, due to its relatively short duration, might not fully represent the anti-tumor activity of NK cells when continuously exposed to immune inhibitory signals in the tumor microenvironment. Therefore, we initiated an *in vivo* tumor model to further investigate the impact of *Prdm1* on NK cell anti-tumor capabilities. B16F10 is a melanoma cell line with low expression of MHC-I, which was susceptible to NK cell killing and usually used to evaluate NK cell anti-tumor capacity (40, 41). The B16F10 cells were intravenously (for lung metastasis) or intrasplenic (for liver metastasis) administrated in the mice. The melanoma nodes were quantified three (intravenous injection) or two (intrasplenic injection) weeks after tumor inoculation. Compared with *Prdm1*^{+/+} mice, deficiency of *Prdm1* resulted in more metastasis nodules in both lung (~ 2-fold) and liver (~ 4-fold) (Figure 2, D and F). Histological analysis further confirmed the increased frequency of metastasis tumor foci in *Prdm1*^{ΔNcr1} mice (Figure 2, E and G).

In agreement with our *in vivo* data, we also observed decreased IFN-γ secretion in *Prdm1*^{ΔNcr1} mice-derived splenic cNK cells, liver cNK cells, and liver ILC1s when stimulated by IL-18 alone or IL-12/IL-18 (Figure 2H; Supplemental Figure3), which indicated that *Prdm1* is required for full activation of cNK cells and ILC1s in the context of IFN-γ production. These data implying that although similar cellular killing capacity was observed between *Prdm1*^{+/+} and *Prdm1*^{ΔNcr1} mice, *Prdm1* is indispensable for NK cell mediated tumor surveillance.

Bulk RNA-seq depicts *Prdm1*-mediated functions in peripheral cNK cells

Bulk RNA sequencing of splenic cNK cells (CD3⁺NK1.1⁺NKp46⁺) was conducted to uncover the molecular mechanisms by which *Prdm1* regulates NK cell anti-tumor immunity (Figure 3A). Differentially expressed genes (DEGs) between *Prdm1*^{+/+} and *Prdm1*^{ΔNcr1} mice were determined

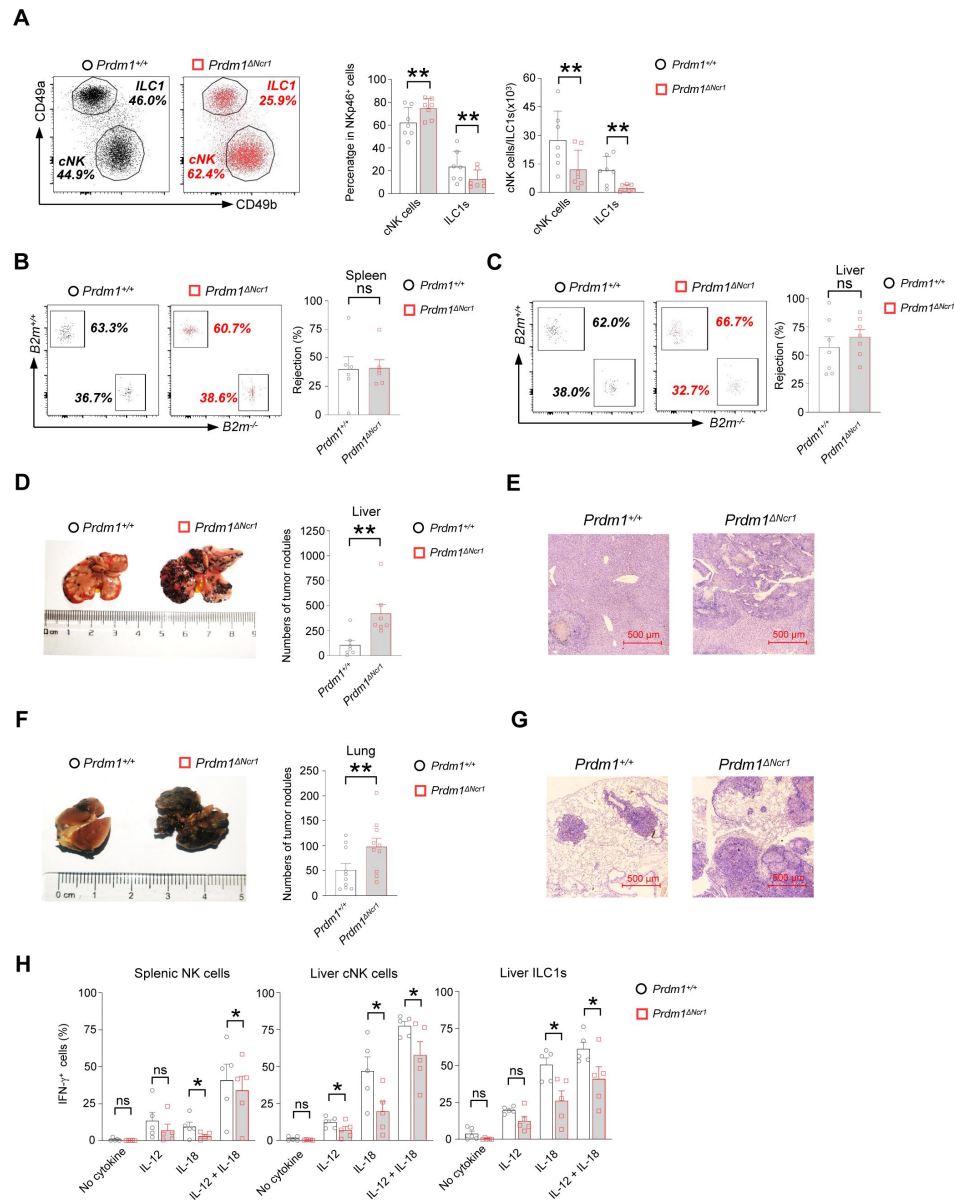


Figure 2.

Prdm1 is required for type I ILCs to control tumor metastasis.

(A) Representative flow cytometric plots (left) of liver cNK cells (CD49a⁺CD49b⁺) and ILC1s (CD49a⁺CD49b⁻) from *Prdm1*^{+/+} and *Prdm1*^{ΔNcr1} mice. The two bar graphs (right) quantitate the percentages and absolute numbers of cells respectively (n=7). (B and C) Splenocytes from *B2m*^{-/-} and *B2m*^{+/+} were labeled with CFDASE and eF670 respectively. Labeled cells were 1:1 mixed and injected i.v. into *Prdm1*^{+/+} and *Prdm1*^{ΔNcr1} recipient mice to evaluate *in vivo* NK cell target-killing ability. Representative flow cytometric plots (left) of transferred cells recovered from recipient mice and percentage (right) of NK cell-specific rejection of donor cells in spleen (n=6) (B) and liver (n=7) (C) between *Prdm1*^{+/+} and *Prdm1*^{ΔNcr1} mice. (D and F) Image (left) and quantification (right) of tumor nodes on the livers (n=7) (D) and lungs (n=10) (F) of *Prdm1*^{+/+} and *Prdm1*^{ΔNcr1} mice at day 14 or 21 after inoculation with B16F10 melanoma cells. (E and G) Histopathological images of liver (E) and lung (G) tissues stained by hematoxylin-eosin to detect tumor metastasis. Red bar indicates 500 μm distance under the microscope. (H) Liver cells and splenocytes were costimulated in the presence or absence of IL-12 and IL-18 for 12 hours. GolgiStop was added 4 hours before intracellular staining of IFN-γ. The graphs showed percentage of IFN-γ⁺ splenic cNK cells, liver cNK cells and ILC1s from *Prdm1*^{+/+} and *Prdm1*^{ΔNcr1} mice (n=5). Data are presented as the mean±SEM and were analyzed by 2-tailed, paired t-test. Differences were evaluated between littermates. Each circle and square on graphs represents an individual mouse; P, P-value; *, P<0.05; **, P<0.01, ns, not significant.

using a criterion of $\log_2$ (fold change) > 0.5 and $P < 0.05$. 445 DEGs were identified out of 17434 protein-coding genes, which consisted of 223 upregulated genes and 222 downregulated genes (Figure 3B).

Gene Ontology (GO) analysis revealed the enrichment of glucuronate metabolism and lymphocyte differentiation in upregulated genes in *Prdm1*^{ΔNcr1} mice derived NK cells (Figure 3C), both of which were associated with cellular growth and development. In contrast, leukocyte mediate cytotoxicity, immune receptor activity, and integrin binding was enriched in the genes which decreased their expression level in *Prdm1*^{ΔNcr1} mice (Figure 3C).

Gene Set Enrichment Analysis (GSEA) showed enriched NF-kappa B signaling pathway and negative regulation of IL-6 production pathway in *Prdm1*-deficient cNK cells (Figure 3, D and E), suggesting the potential targets by *Prdm1* to regulate NK cell function. Increased expression of multiple TFs such as *Junb*, *Batf3*, *Nfkb1*, *Tcf7*, and *Nr4a2* was observed in *Prdm1* knockout cNK cells, suggesting they might be suppressed by *Prdm1* (Figure 3F). Downregulation of granzyme B (*Gzmb*), Perforin (*Prf1*) were observed in *Prdm1* deficient NK cells (Figure 3F), implied decreased anti-tumor ability, which was consistent with increased melanoma metastasis in *Prdm1*^{ΔNcr1} mice (Figure 2D and F). *Cxcr6*, and *Cx3cr1* were also decreased in the NK cells derived from *Prdm1*^{ΔNcr1} mice (Figure 3F), which was considered to play an important role in promoting the egress of NK cells from bone marrow. This might be the reason for the quantity of NK cells increased in bone marrow while decreasing in other peripheral tissues (Figure 1F). As a result of the reduced expression levels of *Cxcr6* and *Cx3cr1*, NK cells were unable to egress from the bone marrow and accumulated therein. Consistent with decreased production of IFN-γ after stimulated by IL-12/IL-18 (Figure 2H), decreased expression of *Il18rap* and *Il12rb2* were observed in *Prdm1*^{ΔNcr1} cNK cells (Figure 3F), implying impaired response to cytokine stimulation.

To confirm the result of RNA-sequencing, the expression of granzyme B and perforin were also analyzed by flow cytometry. Lower granzyme B and perforin production was observed in *Prdm1*-deficient splenic cNK cells, liver cNK cells and ILC1s (Figure 3, G and I; Supplemental Figure 4, A and B). Notably, granzyme B decreased among all of the maturation stages of cNK cells. Probably because of relatively low level of perforin in CD11b⁺CD27⁺ and CD11b⁺CD27⁺ cNK cells, there is not statistical difference between *Prdm1*^{+/+} and *Prdm1*^{ΔNcr1} cNK cells. However, significantly decrease in Perforin was observed in *Prdm1*^{ΔNcr1} mice derived CD11b⁺CD27⁺ NK cells, which have the highest level of Perforin. (Figure 3, H and J; Supplemental Figure 4C). These data implied that *Prdm1* directly regulates NK cells to exert cytotoxicity, not only due to the less mature phenotype in *Prdm1*-deficient NK cells.

scRNA-seq reveals unique properties of two clusters from liver type I ILCs following *Prdm1* knockout

To further investigate the effect of *Prdm1* in liver cNK cells and ILC1s, as well as the changes in the hepatic immune microenvironment caused by the deficiency of *Prdm1* in Type 1 ILCs, single-cell RNA sequencing (scRNA-seq) was performed for CD45⁺ cells sorted from the liver (Figure 3A; Supplemental Figure 5A). Initial quality control revealed high-quality of cell purity, library assembly, and sequencing (Supplemental Figure 5B). 10978 cells passed the quality criteria and were selected for further analysis (6161 from *Prdm1*^{+/+} mice, 4,817 from *Prdm1*^{ΔNcr1} mice). Unsupervised clustering of all sequenced cells based on transcript signatures identified twelve distinct clusters, including B cells, Macrophages, CD4⁺ T cells, CD8⁺ T cells, dendritic cells (DCs), NKT cells, cNK cells, ILC1s, Neutrophils, Monocytes, Basophils, and a small number of undefined cells (Figure 4A). A higher percentage of Macrophages, CD8⁺ T cells, Monocytes, DCs, and a lower percentage of NK cells, ILC1s, CD4⁺ T cells, and NKT cells was observed in *Prdm1*^{ΔNcr1} mice (Supplemental Figure 5C). Liver cNK cells and ILC1s were identified based on the expression of surface markers and TFs (Supplemental Figure 5D). For example, liver cNK cells expressed high

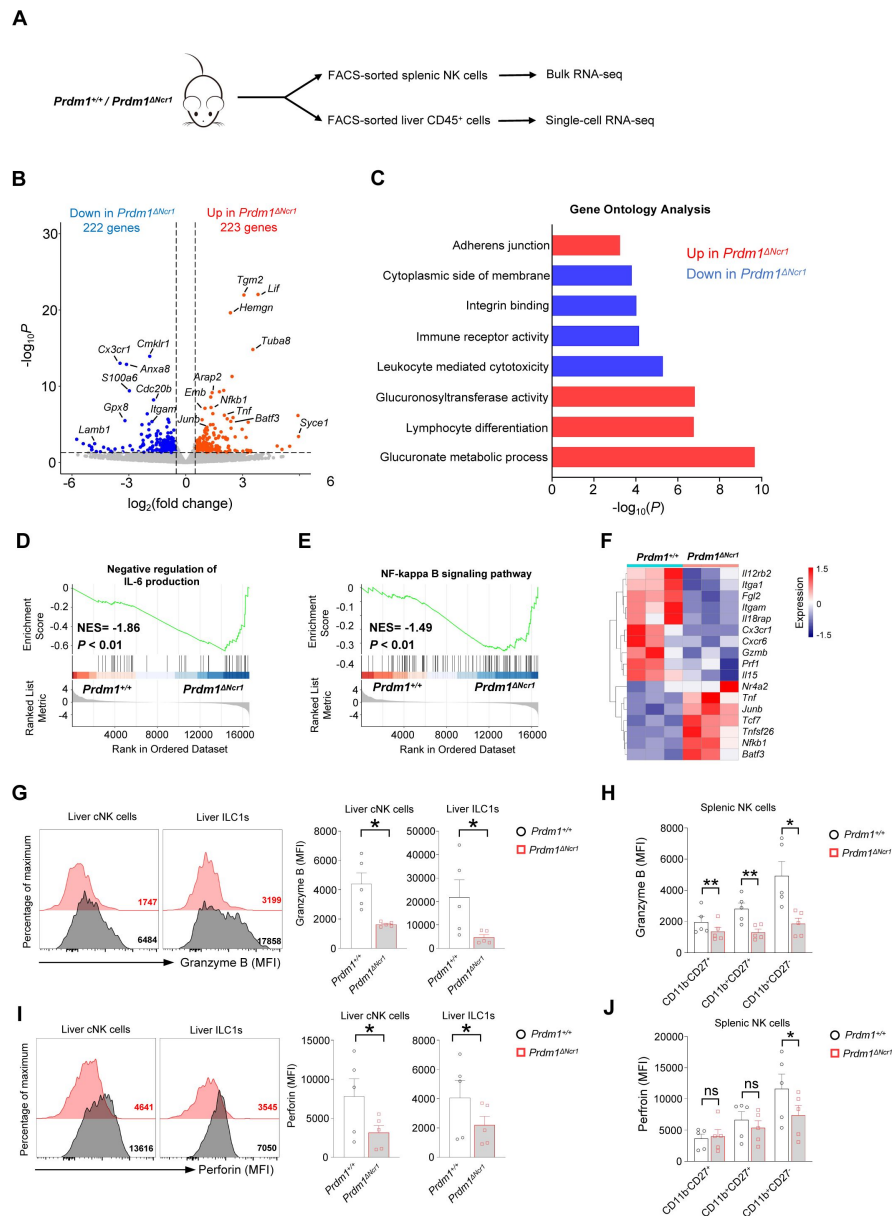


Figure 3.

Bulk RNA-seq depicts *Prdm1*-mediated functions in peripheral cNK cells.

(A) Splenic cNK cells and liver CD45⁺ cells were sorted from *Prdm1*^{+/+} and *Prdm1*^{ΔNcr1} mice using flow cytometry, and prepared for bulk RNA-seq and single-cell RNA-seq analysis. (B) Volcano plot of the bulk RNA-seq differentially expressed genes (log₂(fold change) > 0.5; P < 0.05) in splenic cNK cells between *Prdm1*^{+/+} and *Prdm1*^{ΔNcr1} mice. Upregulated and downregulated genes in *Prdm1*^{ΔNcr1} cells were highlighted in red and blue. (C) Enriched Gene Ontology (GO) terms of DEGs in *Prdm1*^{ΔNcr1} cells compared *Prdm1*^{+/+} cells. The Enrichment gene set in upregulated (red) and downregulated (blue) genes were indicated in different colours. Bar length represents statistical significance. (D and E) Gene Set Enrichment Analysis (GSEA) showing the enrichment of negative regulation of IL-6 production (D) and NF-kappa B signaling pathway (E) of DEGs in *Prdm1*^{ΔNcr1} cells compared *Prdm1*^{+/+} cells. NES, normalized enrichment score. (F) Heatmap of selected genes from DEGs. Shown is z-score transformed expression of DEGs. (G and I) Representative flow cytometric plots (left) and cumulative data (right) showing the relative mean fluorescence intensities (MFIs) of Granzyme B (G) and Perforin (I) in liver cNK cells and ILC1s from *Prdm1*^{+/+} and *Prdm1*^{ΔNcr1} mice (n=5). (H and J) Relative MFIs of Granzyme B (H) and Perforin (J) in splenic NK cells at different stages of maturation was analyzed by flow cytometry (n=5). Data are presented as the mean ± SEM and were analyzed by 2-tailed, paired t-test. Differences were evaluated between littermates. Each circle and square on graphs represents an individual mouse; P, P-value; *, P < 0.05; **, P < 0.01, ns, not significant.

levels of *Itga2* (CD49b) and Eomes, while ILC1s had high levels expression of *Itga1* (CD49a) and *Tnfrsf10* (Supplemental Figure 5E). In liver Type 1 ILCs, increased percentage of cNK cells and decreased percentage of ILC1s were observed in *Prdm1*^{ΔNcr1} mice (**Figure 4, B** and **C**), which was in line with the flow cytometry data (**Figure 2A**).

To better understand the specific function of *Prdm1* played in liver cNK cells and ILC1s, the two subpopulations of liver Type 1 ILCs were further analyzed separately using unsupervised clustering. Liver cNK cells and ILC1s were grouped into three clusters respectively and visualized by Uniform Manifold Approximation and Projection (UMAP) (**Figure 4, D-I**). Based on the cluster specific gene expression signature (Supplemental Figure 6, A and B), the subpopulation of liver cNK cells were referred as “*Prf1*^{hi}”, “*Junb*^{hi}”, and “*Cxcr3*^{hi}” cNK cells (**Figure 4, D-F**).

The *Prf1*^{hi} cNK cell cluster was defined by high expression of cytotoxicity-related genes, including *Ncr1*, *Gzma*, *Gzmb*, *Prf1*, and *Fgl2* (**Figure 4J**; Supplemental Figure 6C), indicating the strong target-killing ability of this cluster. GO analysis further revealed the enrichment signatures of cytotoxicity, response to virus, and lymphocyte mediated immunity in the genes upregulated in *Prf1*^{hi} cNK cell cluster, further confirming the cytotoxic effects of this cluster (**Figure 4K**). While this cluster is present in both *Prdm1*^{ΔNcr1} and *Prdm1*^{+/+} mice, there is a significant reduction in *Prdm1*^{ΔNcr1} mice (**Figure 4F**). Additionally, both *Gzma* and *Gzmb* expression were downregulated in *Prdm1*^{ΔNcr1} mice derived cNK cells and ILC1s compared to those from *Prdm1*^{+/+} mice (**Figure 5A**). These data underscore the crucial role of *Prdm1* in maintaining NK cells with immune effector functions.

The *Junb*^{hi} liver cNK cell cluster distinguished themselves by higher expression of *Junb* compared to other clusters (**Figure 4J**). The expression of *Junb* was also upregulated in *Prdm1*^{ΔNcr1} derived cNK cells and ILC1s (**Figure 5A**). The predominant majority (92.98%) of *Junb*^{hi} liver cNK cells are derived from *Prdm1*^{ΔNcr1} mice, with less than ten percent (7.02%) originating from *Prdm1*^{+/+} mice (**Figure 4F**). Many signal transduction elements, gene expression regulator, and transcriptional factors, such as *Nfkb1a*, *Tnfaip3*, *Nr4a1/2/3*, *Batf3*, *Fos*, *Fosb*, *Tcf7*, and *Kit* were upregulated in the *Junb*^{hi} liver cNK cells (**Figure 4J**; Supplemental Figure 6C). The expression of cytotoxicity related genes, such as *Gzmb* and *Prf1*, in *Junb*^{hi} cluster was also lower than other cNK cells (**Figure 4J**). GO analysis showed that the genes upregulated in *Junb*^{hi} liver cNK cells enriched in cell differentiation, cell activation, and transcriptional regulation (**Figure 4K**). GSEA indicates that the NF-kappa B, IL-17, MAPK, and TNF signaling pathways were upregulated in this clusters (**Figure 5, E-H**). GSEA also showed that mitochondrial related pathways, such as mitochondrial protein, oxidative phosphorylation, and respiratory electron transport chain were suppressed in *Junb*^{hi} cNK cell cluster (Supplemental Figure 6, D-F). Increased proportion of *Junb*^{hi} cluster in *Prdm1* deficient cNK cells suggested impaired anti-tumor activity, which was consistent with more melanoma metastasis in *Prdm1*^{ΔNcr1} mice and lower expression of cytotoxicity-related genes in splenic cNK cells based on bulk RNA-sequencing.

The *Cxcr3*^{hi} cNK cell cluster was characterized with high expression of *Cxcr3*, *Ccr2*, and some genes encoding ribosomal subunits such as *Rps7* (Supplemental Figure 6C). The enrichment of chemokine receptors in the genes upregulated in the *Cxcr3*^{hi} cluster implying a greater likelihood of them being tissue-resident compared with other cNK cell clusters (**Figure 4K**). The significant enrichment of ribosomal subunits and cytoplasmic translation in *Cxcr3*^{hi} cluster (**Figure 4K**) revealed their distinct and active metabolic profile and the capability to mount immune responses. The *Cxcr3*^{hi} cNK cell cluster had a remarkably decreased proportion in *Prdm1*^{ΔNcr1} mice (**Figure 4F**), which emphasized the critical role of *Prdm1* in maintaining this cluster of liver cNK cells. Consistent with the flow cytometry result that showed an increase in the number of NK cells in the bone marrow and a decrease in NK cells in peripheral tissues, bulk RNA sequencing data also found that the expression of *Cx3cr1* and *Cxcr6* decreased in *Prdm1*^{ΔNcr1} cNK

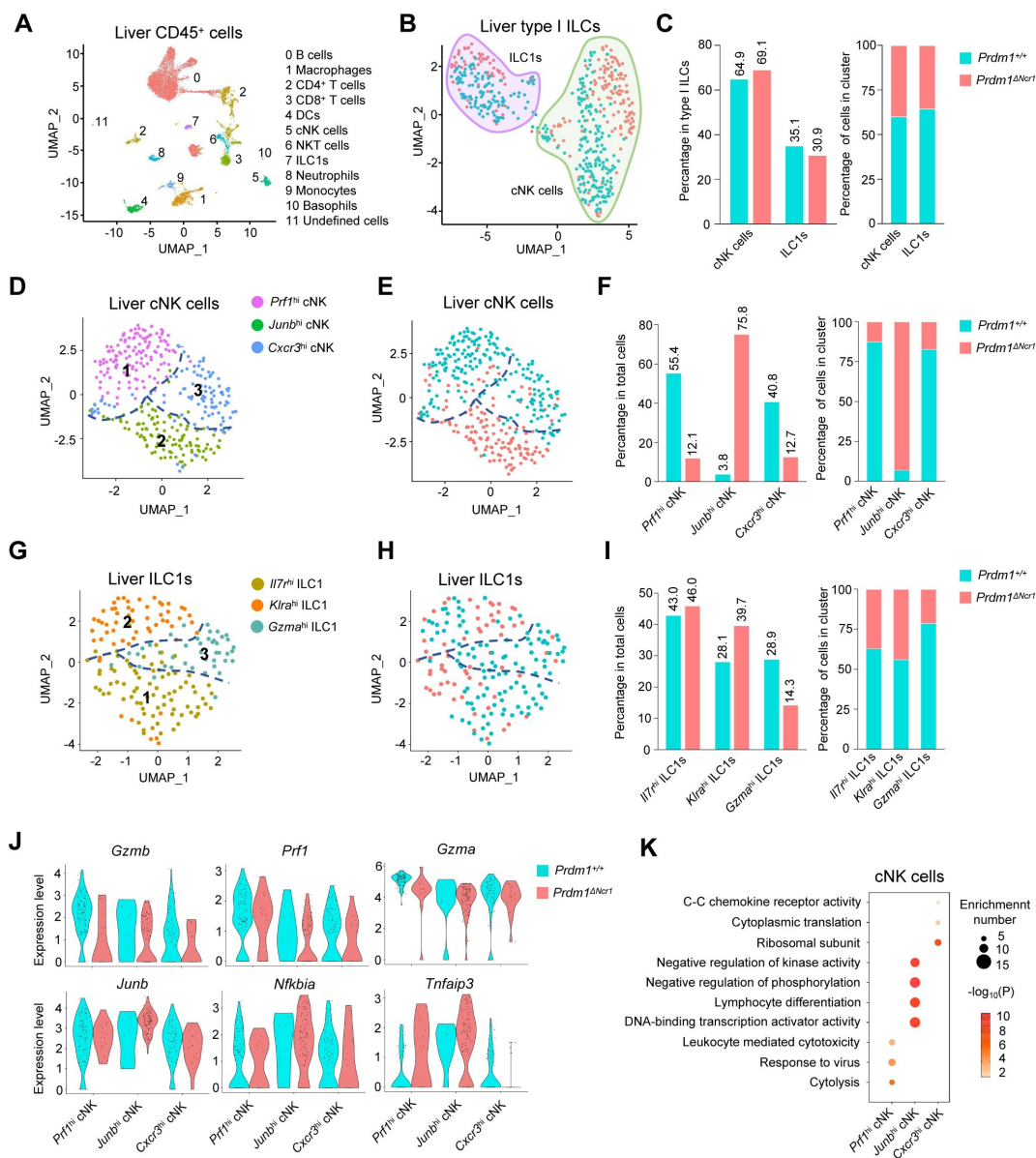


Figure 4.

scRNA-seq reveals unique properties of two clusters from liver type I ILCs following *Prdm1* knockout.

(A) Uniform manifold approximation and projection (UMAP) visualization of liver CD45⁺ cells from *Prdm1*^{+/+} and *Prdm1*^{ΔNcr1} mice. Twelve clusters were defined and indicated by distinct colours. Each dot represents a single cell. (B) UMAP visualization of liver cNK and ILC1 clusters. Cells were colored by origins (*Prdm1*^{+/+}-blue; *Prdm1*^{ΔNcr1}-red). (C) Percentages of cNK cells and ILC1s in type I ILCs (left), and their distribution in each cluster (right). (D-I) UMAP visualization of three different liver cNK (D) and ILC1 (G) clusters from two mouse strains (E and H). Proportions of cNK cells (F) and ILC1s (I) among total cells (cNK cells or ILC1s) (left) and within clusters (right) were calculated respectively. (J) Violin plots showing the normalized expression of select genes in different cNK clusters. (K) Enriched GO term of marker genes in three cNK clusters. Dot size represents enriched gene number, and color intensity represents significance.

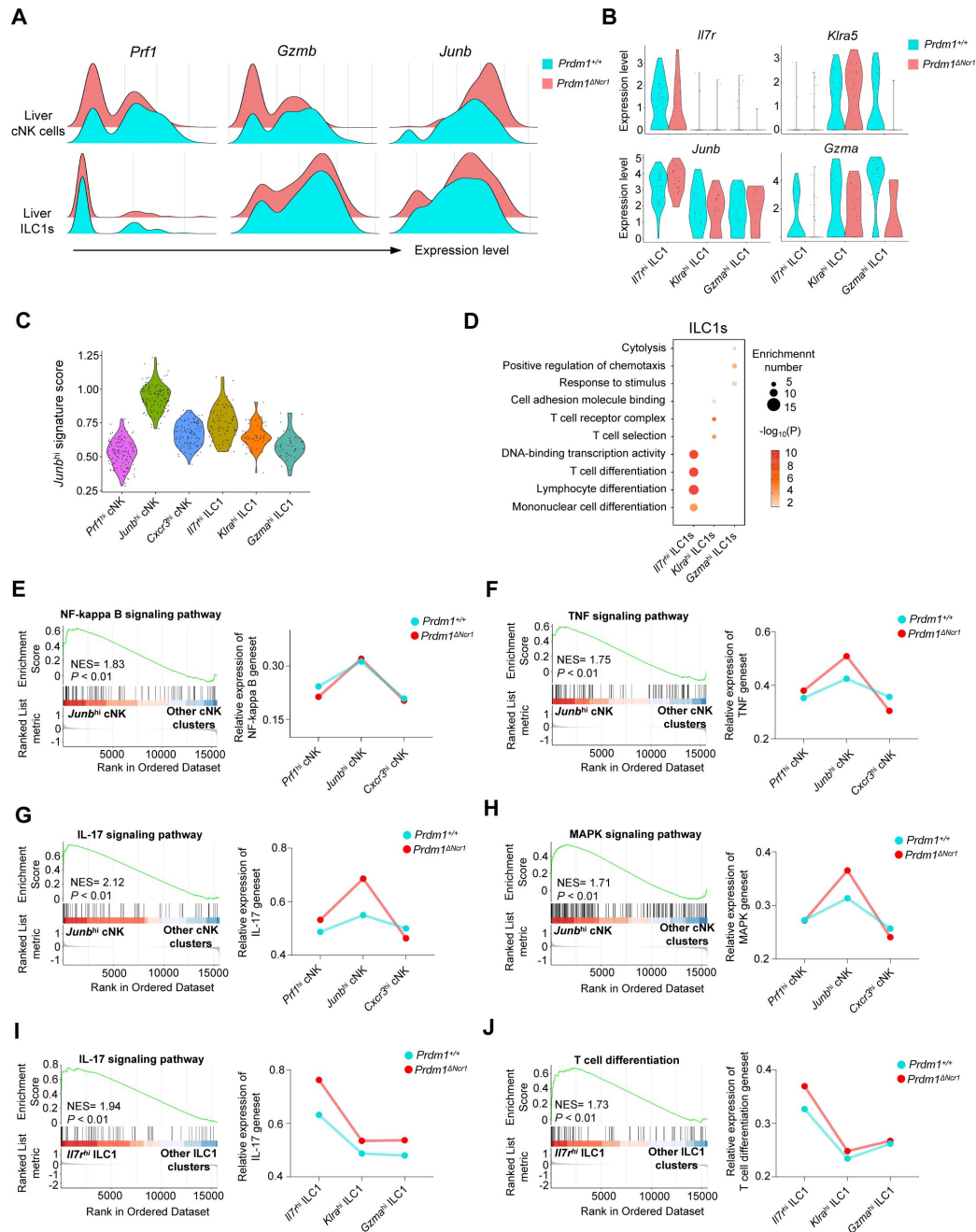


Figure 5.

Junb^{hi} signature is associated with the activation of multiple signaling pathways.

(A) Ridge plots showing the normalized expression of *Gzmb*, *Prf1*, and *Junb* in cNK and ILC1 clusters between *Prdm1*^{+/+} and *Prdm1*^{ΔNcr1} cells. (B) Violin plots showing the normalized expression of select genes in different ILC1 clusters. (C) Violin plot showing the *Junb*^{hi} signature score for cNK cell and ILC1 clusters, calculated using the signature genes of *Junb*^{hi} cNK cluster. (D) Enriched GO term of marker genes in three ILC1 clusters. Dot size represents enriched gene number, and color intensity represents significance. (E-J) GSEA plots (left) depicting the enrichment of NF-kappa B (E), TNF (F), IL-17 (G), and MAPK (H) signaling pathway in *Junb*^{hi} cNK cluster compared with clusters of *Prf1*^{hi} and *Cxcr3*^{hi} cNK cells, and the enrichment of IL-17 signaling pathway (I) and T cell differentiation (J) in *Il7r*^{hi} ILC1 cluster compared with clusters of *Klra*^{hi} and *Gzma*^{hi} ILC1s. Right panel showed dynamic relative expression of the given gene sets from cluster1 to cluster3 between *Prdm1*^{+/+} and *Prdm1*^{ΔNcr1}. Dots represent the average expression of given gene set in each cell, which was calculated through the sum of normalized expression of each individual gene within the designated gene set in every single cell. NES, normalized enrichment score.

cells. These findings supported the hypothesis that *Prdm1*, through regulating chemokine receptor expression levels, influenced the distribution of NK cells in the bone marrow and peripheral tissues, particularly within the liver tissue.

Three clusters of ILC1s were identified from liver ILC1s (**Figure 4, G-I**), comprising “*Il7r*^{hi}”, “*Klra*^{hi}”, and “*Gzma*^{hi}” ILC1s. The first two clusters of ILC1s were characterized by higher expression of *Il7r* (CD127) and *Klra5* separately, while the *Gzma*^{hi} ILC1 cluster was identified by elevated expression of both *Gzma* and *Gzmb* (Supplemental Figure 7, A-C). The *Il7r*^{hi} ILC1s cluster was distinguishable from other ILC1 clusters by its high expression of *Il7r*, IL18RA (*Il18r1*), and IFN-γ (*Ifng*) (**Figure 5B**; Supplemental Figure 7C). The high expression of *Il18r1* and *Ifng* in *Il7r*^{hi} ILC1s indicated this cluster of cells was highly responsive to IL-18. Module scores, calculated based on the expression of feature genes within the *Junb*^{hi} cNK cell cluster, revealed a comparable *Junb*^{hi} signature expression pattern within the *Il7r*^{hi} ILC1 cluster (**Figure 5C**). GO analysis and GSEA showed that IL-17, NF-kappa B, TNF, MAPK signaling pathway and T cell differentiation were activated in the *Il7r*^{hi} ILC1 cluster (**Figure 5D, I** and **J**; Supplemental Figure 7D-F). Considering the close relationship between IL-17 mediated immunity response and ILC3 (42), it is plausible that the observed heterogeneity within this cluster may be attributed, at least in part, to potential plasticity between ILC1 and ILC3 subsets. These two clusters appear reminiscent of an intermediate stage in the conversion process from cNK cells to ILC1s, as previously described (23, 43). There was no significant difference in the proportion of this cluster between WT and in *Prdm1*^{ΔNcr1} mice.

The second liver ILC1 cluster, characterized by high expression of Ly49E (*Klra5*) and Ly49G (*Klra7*), was designated as the *Klra*^{hi} ILC1 cluster (**Figure 5B**). Notably, there was an elevated proportion of *Klra*^{hi} ILC1s in *Prdm1*^{ΔNcr1} ILC1s (39.7%) compared to *Prdm1*^{+/+} ILC1s (28.1%). Liver Ly49E⁺ ILC1s have been identified as possessing greater cytotoxic potential and a more robust viral response compared to liver Ly49E⁻ ILC1s (44). The *Klra*^{hi} cluster exhibited notably high expression of *Ccl5* (Supplemental Figure 7C). Previous research has underscored the pivotal role of CCL5, produced by both cNK cells and ILC1s, in facilitating the accumulation of DCs within the tumor microenvironment, thereby impeding tumor immune evasion, as highlighted in studies (32, 45). The expression of *Ccl5* was reduced in the *Klra*^{hi} cluster of *Prdm1*^{ΔNcr1} ILC1s compared to *Prdm1*^{+/+} ILC1s (Supplemental Figure 7C). This decrease in *Ccl5* expression could potentially have a detrimental impact on the ability of *Klra*^{hi} ILC1s to develop a connection between innate and adaptive immune responses.

The *Gzma*^{hi} ILC1 cluster was identified according to the high expression of *Gzma*, *Gzmb*, and *Fgl2* (Supplemental Figure 7C), compared to other ILC1 clusters. GO analysis also revealed the enrichment in cytolysis and stimulus-response capacity (**Figure 5D**) of the *Gzma*^{hi} ILC1 cluster. Consistent with the *Prf1*^{hi} cNK cell cluster, the proportion of the *Gzma*^{hi} cluster among liver ILC1s exhibited a considerable reduction in *Prdm1*^{ΔNcr1} mice compared to *Prdm1*^{+/+} mice, and the expression of *Gzma* also downregulated in *Prdm1*^{ΔNcr1} mice (**Figure 5B**). Previous reports showed that Gzma⁺ ILC1 constituted the main population of liver ILC1s at birth, with the potential target-killing ability (46, 47).

***Prdm1* facilitates the intercellular communication between liver Type 1 ILCs and macrophages**

The reciprocal crosstalk between Type 1 ILCs and macrophages plays a critical role in maintaining liver immune homeostasis and anti-cancer immune surveillance. Based on the scRNA-seq data, four clusters were generated from liver macrophages according to the transcriptional profile. They were labeled as “*Ly6c2*^{hi}”, “*Cxcl2*^{hi}”, “*Ear2*^{hi}”, and “*C1q*^{hi}” macrophages (Mac) (**Figure 6A**, Supplemental Figure 8, A-D).

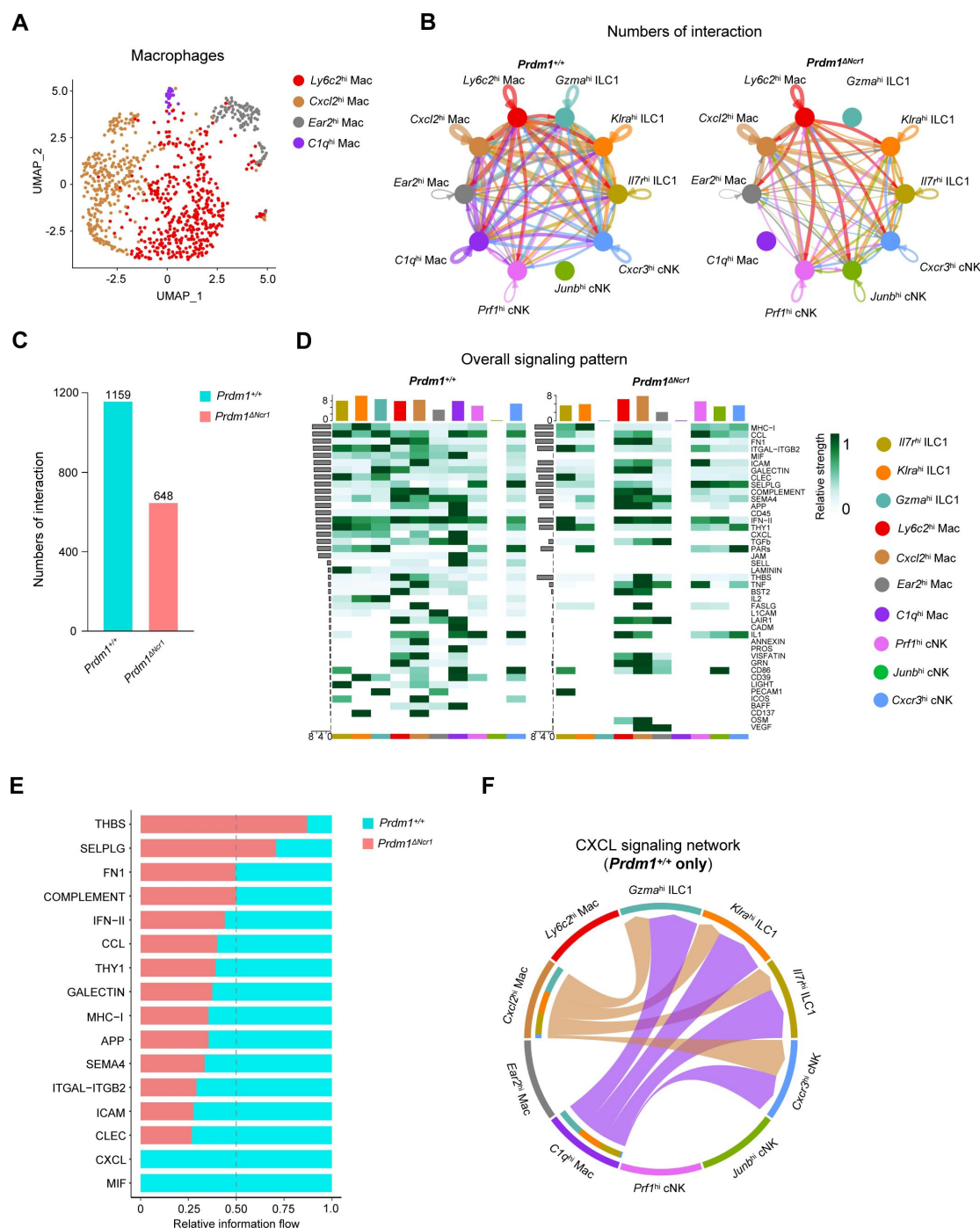


Figure 6.

***Prdm1* facilitates the intercellular communication between liver type I ILCs and macrophages.**

(A) UMAP visualization of macrophages (Mac) cluster. Four cell subsets were defined. (B and C) Circle plots (B) and summary data (C) illustrating the significant enriched ligand-receptor pairs among cluster of liver cNK cells, ILC1s, and macrophages from *Prdm1*^{+/+} (left) and *Prdm1*^{ΔNcr1} (right) cells. The thickness of the line indicates the number of enriched pairs, and the arrow reflects the direction of the interaction. (D) Heatmap of overall signaling pattern recognized from ligand-receptor pairs, which contained the sum of signaling from the sender and target cells. (E) Bar graphs showing the information flow in selected active signaling patterns between *Prdm1*^{+/+} and *Prdm1*^{ΔNcr1} cells. Relative information flow was calculated as the sum of the communication probability in given signaling patterns. (F) Chord plot of the CXCL signaling interaction network among cluster of liver cNK cells, ILC1s, and macrophages in *Prdm1*^{+/+} cells.

High-resolution interactions among liver cNK cells, ILC1s, and macrophages were established and compared between *Prdm1*^{+/+} and *Prdm1*^{ΔNcr1} mice using the CellChat program (48 [↗](#)). Interactions between ILC1s and macrophages were higher than that between cNK cells and macrophages (Supplemental Figure 9, A and B). There is a significant decrease in the interaction number between liver Type 1 ILCs (including cNK cells and ILC1s) and macrophages (~2 fold) in *Prdm1*^{ΔNcr1} mice (Figure 6, B [↗](#) and C [↗](#)). A reduction in the interaction of ligand-receptor, such as Mif-CD74, Cxcl16-Cxcr6, and Cxcl10-Cxcr3 was observed in *Prdm1*^{ΔNcr1} mice compared to *Prdm1*^{+/+} mice (Supplemental Figure 10). Liver cNK cells and ILC1s contributed equally to macrophages through most pathways, while CXCL chemokine ligand (CXCL), Thy-1 cell surface antigen (THY1), and C-type lectin (CLEC) pathways participated more in ILC1-Mac interaction than cNK-Macrophage interaction (Supplemental Figure 9, C and D). Compared to *Prdm1*^{+/+} mice, the information flow of CXCL and macrophage migration inhibitory factor (MIF) pathways significantly decreased in *Prdm1*^{ΔNcr1} mice (Figure 6, D [↗](#) and E [↗](#)). These pathways play a crucial role in facilitating macrophage migration. The pathway Thrombospondin (THBS) was upregulated in *Prdm1*^{ΔNcr1}, which was consistent with the higher expression of *Thbs1* in *Prdm1*^{ΔNcr1} macrophages (Supplemental Figure 8C). The CXCL signaling was sent from *Cxcl2*^{hi} macrophage and *C1q*^{hi} macrophage, targeting all ILC1 clusters and *Cxcr3*^{hi} cNK cell clusters (Figure 6F [↗](#)). Of note, although the population of *Cxcl2*^{hi} macrophage primarily comprised cells from *Prdm1*^{ΔNcr1} mice, the interaction within the CXCL pathway between macrophages and Type 1 ILCs was only observed in *Prdm1*^{+/+} sample (Figure 6F [↗](#)). These changes could be linked to a decreased population of ILC1s and *Cxcr3*^{hi} cNK cell cluster in *Prdm1*^{ΔNcr1} mice, implying that the homeostasis of *Cxcl2*^{hi} macrophages required sufficient signals from cNK cells and ILC1s. The impaired CXCL-CXCR interactions might subsequently lead to reduced recruitment and activation of Type 1 ILCs and macrophages within the tumor microenvironment.

***Prdm1* safeguards Type 1 ILCs from exhaustion-like phenotypes in the tumor microenvironment**

The suppression of mitochondrial related pathways in *Junb*^{hi} cNK cell cluster, along with a significant increase of this cNK cell cluster in *Prdm1*^{ΔNcr1} mice, encouraged us to explore mitochondrial function through flow cytometry. MitoTracker, MitoSOX, and Tetramethylrhodamine methyl ester (TMRM) were used to assess the mitochondrial mass, superoxide production, and mitochondrial membrane potential. A substantial decrease in mean fluorescent intensities (MFIs) of MitoTracker was observed in *Prdm1*^{ΔNcr1} splenic cNK cells, liver cNK cells, and liver ILC1s when compared to their *Prdm1*^{+/+} counterparts (Figure 6A [↗](#)). This observation aligns with the enrichment of downregulated genes from *Prdm1* deficient sample in mitochondrial related pathway, as revealed by RNA sequencing data (Supplemental Figure 6, D-F). There was no significant difference in MitoSOX and TMRM between *Prdm1*^{ΔNcr1} and *Prdm1*^{+/+} mice (Figure 7, B [↗](#) and C [↗](#)), which suggested that the ATP synthesise capacity was minimally affected by *Prdm1*.

IFN-γ is a critical cytokine for NK cells mediated cancer surveillance(49 [↗](#), 50 [↗](#)) and impaired production of IFN-γ was considered as a key hallmark of exhausted NK cells (51 [↗](#), 52 [↗](#)). To evaluate the IFN-γ secreting capacity of liver cNK cells and ILC1s in tumor microenvironment, B16F10 tumor cells were inoculated to the liver via splenic injection and the IFN-γ levels in response to stimulation of IL-12 and/or IL-18 were assessed by flow cytometry (Supplemental Figure 11). Compared with *Prdm1*^{+/+}, significant decreased of IFN-γ were observed in *Prdm1*^{ΔNcr1} liver cNK cells and ILC1s under the combine stimulation of IL-12/IL-18 (Figure 7D [↗](#)), which was more remarkable in liver ILC1s. Similar trends were observed when IL-12 or IL-18 was used alone, although only liver ILC1s showed a significant decrease in response to IL-18 stimulation (Figure 7D [↗](#)). These findings were consistent with the heavy tumor burden observed in *Prdm1*^{ΔNcr1} mice.

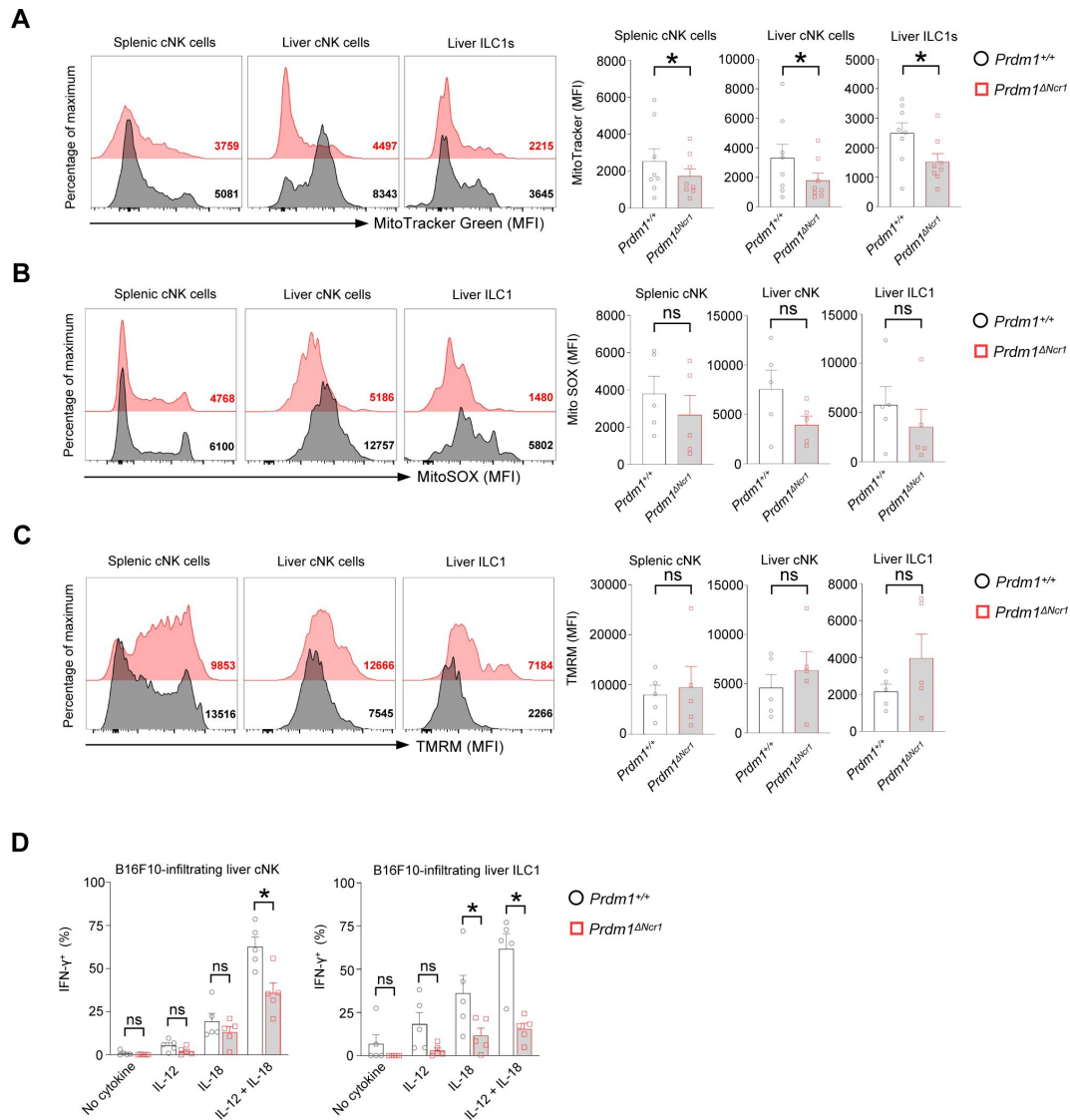


Figure 7.

***Prdm1* safeguards type I ILCs from exhaustion-like phenotypes in the tumor microenvironment.**

(A-C) The Mitochondrial mass (MitoTracker Green staining; n=8) (A), Mitochondrial ROS (MitoSOX staining; n=5) (B), and Mitochondrial membrane potential (TMRM staining; n=5) (C) of splenic cNK cells, liver cNK cells and ILC1s were analyzed by flow cytometry. Representative flow cytometric plots (left) and cumulative data (right) showing the relative mean fluorescence intensities (MFIs) of each group. (D) Percentages of IFN-γ⁺ liver cNK cells and ILC1s from *Prdm1*^{+/+} and *Prdm1*^{ΔNcr1} tumor-bearing mice at day 14 after inoculation with B16F10 melanoma cells via intrasplenic injection (n=5). Data are presented as the mean±SEM and were analyzed by 2-tailed, paired t-test. Differences were evaluated between littermates. Each circle and square on graphs represents an individual mouse; P, P-value; *, P<0.05; **, P<0.01, ns, not significant.

Discussion

Using *Ncr1*-driven conditional knockout transgenic mice, which specifically delete *Prdm1* in Type 1 ILCs, we not only validated *Prdm1*'s positive regulation of NK cell maturation, but also demonstrated its indispensable role in NK cell anti-tumor activity. The compromised mitochondrial function and reduced IFN- γ , granzyme B, and perforin production appear to be potential contributing factors to *Prdm1*-mediated cancer surveillance. Decreased expression of *Cx3cr1* and *Cxcr6* was observed in *Prdm1* ^{Δ *Ncr1*} splenic cNK cells (**Figure 3F**), both of which are essential for NK cells egressing from bone marrow (53, 54). The quantity of cNK cells increased exclusively in the bone marrow, with reductions observed in all other tissues (**Figure 1F**). This indicated that *Prdm1* might regulate chemokine receptors to facilitate the egression of cNK cells from the bone marrow to peripheral tissues. In addition, higher expression of *Cxcr6* compared to cNK cells is also a key factor for liver tissue residence of ILC1s (55). Decreased expression of *Cxcr6* in *Prdm1* deficient Type 1 ILCs may also contribute to the balance shift towards cNK cells.

scRNA sequencing analysis reveals that both liver cNK cells and ILC1s can be further divided into three subgroups based on their gene expression patterns. Within the cNK cells, the abundance of *Junb*^{hi} cluster substantially increased in *Prdm1* ^{Δ *Ncr1*} mice (75.8%). Notably, these cells account for less than 5% of WT cNK cells. High *Junb* expression stood out as a prominent feature of this subgroup. *Junb* is a crucial transcriptional factor for the cytotoxic function of CD8⁺ T cells and NK cells. However, excessive *Junb* expression has been found to promote T cell exhaustion (56). Our previous study showed that as NK cells mature, the expression level of *Prdm1* increased while the expression level of *Junb* gradually decreased (57). Our data demonstrated that in NK cells, the expression level of *Junb* significantly increases upon the deletion of *Prdm1*, indicating that *Junb* expression is suppressed by *Prdm1*. As *Junb* expression decreases with NK cell maturation, and it is inhibited by the gradually increasing *Prdm1* during maturation. This implies that constraining *Junb* expression is likely a fundamental prerequisite for NK cell maturation. However, the precise mechanism by which *Prdm1* downregulates *Junb* in NK cells still needs further research. Furthermore, *Junb*^{hi} NK cells exhibit lower expression levels of cytotoxic genes and reduced mitochondrial-related signaling pathways. Mitochondrial fragmentation has also been confirmed to be closely associated with NK cell exhaustion (58). While our current data is not sufficient to definitively classify these cells as exhausted NK cells, it supports that this particular group of NK cells demonstrates an exhausted phenotype. The significant increase in this cell population following *Prdm1* knockout in NK cells may potentially be one of the reasons why *Prdm1* ^{Δ *Ncr1*} mice lose their tumor-killing capacity.

Our scRNA-seq data revealed that *Prdm1* plays distinct roles in regulating cNK cells and ILC1s despite being required for both lineages. Specifically, *Prdm1* appears to be more involved in promoting the resistance against exhaustion in cNK cells, whereas in ILC1s, it may play a role in the plasticity between ILC1s and ILC3s. In both our previous study and a study by Colonna et al. (57, 59), it was demonstrated that *Smad4*, a transcriptional factor involved in TGF- β signal pathway, upregulated *Prdm1* in NK cells and depletion of *Smad4* resulted in a decreased ratio of NK cells to ILC1s in the liver. However, knocking out *Prdm1* in *Ncr1*⁺ cells increased the ratio of NK cells to ILC1s in liver type 1 ILCs (**Figure 2A**). These findings suggest the possibility of a *Smad4*-independent pathway through which *Prdm1* promotes the maintenance of ILC1s, or that *Prdm1* plays a more significant role in maintaining ILC1s compared to its role in NK cells.

Chronic inflammation is a crucial factor in promoting tumorigenesis, and macrophages play a significant role in this process. Macrophages interact with both cNK cells and ILC1s. However, the TFs that regulate these interactions are poorly understood. Fortunately, recent advances in scRNA-seq technology and the CellChat software tool (48) have allowed us to gain a better understanding of the *Prdm1* signaling pathway in Type 1 ILCs and its impact on macrophage function at the transcriptional level. It is worth noting that normal ILC1-macrophage interactions

are more prevalent than those with NK cells (Supplemental Figure 9, A and B). Moreover, the finding of increased liver metastasis in *Prdm1*^{ΔNcr1} mice indicated the crucial role of *Prdm1* in maintaining homeostasis between Type 1 ILCs and macrophages. Specifically, *Prdm1* may be critical in preventing the overactivation of macrophages that can lead to cancer development.

Methods

Mice

Prdm1^{fl/fl} mice were purchased from The Jackson Laboratory. *Ncr1-iCre* and *B2m*^{-/-} mice were purchased from Shanghai Model Organisms Center, Inc.. Six- to twelve-week-old littermates were used for the experiment.

Experimental metastasis model

For lung metastasis model, 0.3×10^6 B16F10 cells were intravenous injected into mice. Three weeks later, mice were euthanized for analysis. For liver metastasis model, mice were inoculation with 0.5×10^6 B16F10 via intrasplenic injection. Three weeks later, mice were euthanized for analysis. Lung and liver from tumor-bearing mice were fixed in 10% formalin and embedded in paraffin. Sections were stained with H&E.

In vivo cytotoxicity assay

Donor splenocytes harvested from B2m deficient (*B2m*^{-/-}) mice were labeled with 5 μM CFDA-SE. Donor splenocytes harvested from B2m-adequate (*B2m*^{+/+}) mice were labeled with 5 μM eF670. Labeled splenocytes from two mouse strains were mixed in a 1:1 ratio, and 1×10^7 cells in total were injected i.v. into *Prdm1*^{+/+} and *Prdm1*^{ΔNcr1} mice. One day after administration, spleen and liver cells were isolated from recipient mice, and the population of labeled cells was analyzed by flow cytometry. Rejection % was quantified according to the following formula:

$$[1 - (\%B2m^{+/+} / \%B2m^{-/-})_{Prdm1^{+/+} \text{ or } Prdm1^{\Delta Ncr1}}] \times 100\%.$$

Flow cytometry

Flow cytometry and cell sorting were performed with a Cytoflex S/SRT (Beckman Coulter). Data were analyzed using FlowJo software. The following antibodies were used (all purchased from BioLegend unless otherwise indicated): CD45-PE-Cy7 (catalog 103114, clone 30-F11); CD3ε-PerCP-Cy5.5 (catalog 100327, clone 145-2C11); NK1.1-BV421 (catalog 108741, clone PK136); CD335-AF647 (catalog 560755, clone 29A1.4, BD Bioscience); CD49a-PE (catalog 562115, clone Ha31/8, BD Bioscience); CD49b-FITC (catalog 108906, clone DX5); CD27-BV510 (catalog 124229, clone LG.3A10); CD11b-AF700 (catalog 101222, clone M1/70); KLRG1-APC (catalog 561620, clone 2F1, BD Bioscience); CD49a-BV421 (catalog 740046, clone Ha31/8, BD Bioscience); IFN-γ-PE (catalog 505807, clone XMGI.2); Granzyme B (catalog 372207, clone QA16A02); Perforin (catalog 154305, clone S16009A); CD49b-APC-Cy7 (catalog 108919, clone DX5); IgG-PE (catalog 402203, clone 27-35); CD335-BV510 (catalog 137623, clone 29A1.4); TIGIT-PE (catalog 622205, clone A17200C); CD3ε-APC-Cy7 (catalog 100330, clone 145-2C11). For mitochondrial metabolic assay, fresh cells were incubated in 37°C media for 30min with 100 nM MitoTracker Green (catalog M7514, Invitrogen), 100 nM TMRM (catalog T668; Invitrogen), and 10 uM MitoSOX Red (catalog M36008; Invitrogen), respectively. Surface-stained after washing with PBS and then detected by flow cytometry. For intracellular IFN-γ staining, Cells freshly obtained from liver and spleen were stimulated 12 hours with or without cytokine. GolgiStop (BD Biosciences) was added 4 hours before intracellular staining.

Bulk RNA sequencing

Total RNA from FACS sorted splenic NK cells of *Prdm1*^{+/+} and *Prdm1* ^{Δ Ncr1} mice was extracted by TRIzol reagent (Invitrogen), then reverse transcribed into cDNA. Library construction was prepared using Illumina mRNA Library kit, and sequencing was performed by the BGISEQ-500. Standard methods were used to analyze the RNA-seq data, including aligning the reads to the genome by HISAT2 (v2.1.0) (60), and gene expression values (Counts) were calculated using RSEM (v1.3.1) (61). DEGs were identified using DESeq2 (v1.4.5) (62) with a cutoff of $\log_2$ (fold change) > 0.5 and P < 0.05. The “clusterProfiler” package (v4.4.4) (63) and gene sets from molecular signatures database (MSigDB) were used for GSEA and GO analysis. The heatmap was plotted using the “Pheatmap” package (v1.0.12).

Single-cell RNA sequencing

FACS-sorted liver CD45⁺ cells with more than 80% cell viability were used for library preparation. Each sample contained cells from three *Prdm1*^{+/+} or *Prdm1* ^{Δ Ncr1} mice. Gel Bead-in-Emulsions (GEMs) were generated using the 10X Genomics Chromium system, which combines Master Mix, Single Cell 3' v3.1 Gel Beads, and Partitioning Oil with single cells. GEMs were mixed with cell lysate and reverse transcription reagent to produce full-length cDNA. After incubation, the GEMs were broken, and recovered cDNA were amplified via PCR. Fragmentation, End repair, A-tailing, and Adaptor Ligation were performed to obtain final libraries, which contain P5 and P7 sequences. The 3' library was sequenced on Novaseq 6000 with approximately 50k read pairs/cell sequencing depth. The “Seurat” R package (v4.2.0) (64) was used for data analysis. Initial quality control was performed to filter out the low-quality cells and cell doublets. Cells with 200-5500 expressed genes and no more than 10% mitochondrial genes were considered for high-quality. Doublets were filtered with the “scDblFinder” Package (v1.10.0) (65). After quality control, we totally recovered 6161 cells and 4817 cells from *Prdm1*^{+/+} and *Prdm1* ^{Δ Ncr1} mice, respectively. Principal component analysis (PCA) was used for cluster analysis. The first 15 PCs were picked for clustering and further visualized by UMAP. Clusters-specific marker was defined using the “FindAllMarkers” function, and clusters were manually annotated based on the top 30 or 15 markers. The “clusterProfiler” package and gene set from MSigDB were used for GSEA and GO analysis. “CellChat” package (v1.4.0) (48) was utilized to predict the cell-to-cell communication from scRNA-seq data.

TCGA datasets assay

The normalized gene expression and survival datasets of cancer patients collected in The Cancer Genome Atlas (TCGA) were downloaded from UCSC Xena (<http://xena.ucsc.edu/>) (66). The mean expression of NK cell-associated genes (*NCR1*, *NCR3*, *PRF1*, *KLRB1*, *CD160*) was defined as NK signature according to the previous study (32). NK cell-associated genes, together with *PRDM1*, constitute the *NK-PRDM1* signature in this study. The mean expression of per genes was ordered from high-to-low and plotted by heatmap using the “Pheatmap” package. The overall survival of patients in the high and low quartiles expression of *NK-PRDM1* signature was selected for analysis. Kaplan-Meier curves were plotted by GraphPad Prism.

Statistics

For experiment results, two-tailed t tests were used to measure the continuous and normally distributed between the two independent groups. Paired t-tests were used to determine the statistical significance between two paired groups. Log-rank tests were used to compare the overall survival distribution between the two groups of patients. A P value less than 0.05 was considered significant and data were presented as mean $\pm$ SEM.

Study approval

All animal experiments were approved by The Tianjin University Animal Care and Use Committee (protocol 00000000202010100023). No human subjects were performed in this study.

Author contributions

Jitian He, and Youwei Wang designed experiments, performed experiments, analyzed the data and wrote the manuscript. Jitian He, Le Gao, Peiying Wang, Yiran Zheng, and Yumo Zhang performed experiments. Wing Keung Chan, Jiming Wang, Huaiyong Chen, and Zhouxin Yang analyzed the data, interpreted results and reviewed the manuscript.

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Declaration of interests

The authors have declared that no conflict of interest exists.

Supplemental Figure 1. High expression of *PRDM1*-NK signature predicts better overall survival of cancer patients.

(A and D) Correlation between the average expression of NK cell-associated genes (*NCR1*, *NCR3*, *PRF1*, *CD160*, *KLRB1*) and *PRDM1* in skin cutaneous melanoma (SKCM, n=454) **(A)** and lung adenocarcinoma **(D)** (LUAD, n=497) patients from TCGA datasets. **(B and E)** Heatmap of the ordered, z-score normalized expression values for *PRDM1* and NK cell-associated genes in skin cutaneous melanoma **(B)** and lung cancer **(E)** patients. The top and bottom quartiles of all patient samples are indicated. **(C and F)** Prognostic value of the *NK-PRDM1* signature for overall survival of skin cutaneous melanoma **(C)** and lung cancer **(F)** patients comparing high and low quartiles. *P*, *P*-value; *r*, pearson correlation coefficient.

Supplemental Figure 2. *Prdm1* plays an essential role in type I ILCs homeostasis and maturation.

(A) Representative flow cytometric plots (left) and quantification (right) of the proportion and absolute number of CD3⁺NKp46⁺ cells among lymphocytes in blood, spleen, and lymph nodes (n=8). **(B)** Representative flow cytometric plots (left) of the CD11b and CD27 expression within CD3⁺NK1.1⁺NKp46⁺ cells in blood, spleen, and lymph nodes (n=7). Right panel showed the percentage of distinct stages of NK cells. **(C and D)** Representative flow cytometry analyses **(C)** and quantification **(D)** of KLRG1 expression in CD3⁺NK1.1⁺NKp46⁺ cells in blood, bone marrow, lung, liver, spleen, and lymph nodes (n=5). Data are presented as the mean±SEM and were analyzed by 2-tailed, paired t-test. Differences were evaluated between littermates. Each circle and square on graphs represents an individual mouse; *P*, *P*-value; *, *P*<0.05; **, *P*<0.01, ns, not significant.

Supplemental Figure 3. *Prdm1* affects the IFN- γ secretion ability of type I ILCs.

(A) Representative flow cytometric plot showing the frequency of IFN- γ ⁺ cNK cells and ILC1s in spleen and liver between *Prdm1*^{+/+} and *Prdm1* ^{Δ Ncr1} mice (n=5). Liver cells and splenocytes were costimulated in the presence or absence of IL-12 and IL-18 for 12 hours. GolgiStop was added 4 hours before intracellular staining of IFN- γ . Data are presented as the mean $\pm$ SEM and were analyzed by 2-tailed, paired t-test. Differences were evaluated between littermates. Each circle and square on graphs represents an individual mouse; *P*, *P*-value; *, *P*<0.05; **, *P*<0.01, ns, not significant.

Supplemental Figure 4. *Prdm1* deficiency impairs production of Granzyme B and Perforin in type I ILC.

(A and B) Representative flow cytometric plot (left) and summary data (right) showing the relative mean fluorescence intensities (MFIs) of Granzyme B (A) and Perforin (B) in splenic NK cells from *Prdm1*^{+/+} and *Prdm1* ^{Δ Ncr1} mice (n=5). (C) Splenic NK cells were divided into different stages according to the expression of maturation markers CD11b and CD27. Representative flow cytometric plot showing the MFIs of Granzyme B and Perforin in evaluated between littermates. Data are presented as the mean $\pm$ SEM and were analyzed by 2-tailed, paired t-test. Differences were evaluated between littermates. Each circle and square on graphs represents an individual mouse; *P*, *P*-value; *r*, pearson correlation coefficient; *, *P*<0.05; **, *P*<0.01, ns, not significant.

Supplemental Figure 5. scRNA-seq identified subsets of liver CD45⁺ cell from *Prdm1*^{+/+} and *Prdm1* ^{Δ Ncr1} mice.

(A) Liver CD45⁺ cells from *Prdm1*^{+/+} and *Prdm1* ^{Δ Ncr1} mice were FACS-sorted for scRNA-seq. (B) Quality control of the scRNA-seq data. Violin plot showing the nFeature_RNA, nCount_RNA, and percentage of mitochondrial genes of sequencing cells. nFeature_RNA represents the number of genes detected in each cell, and nCount_RNA represents the total number of molecules detected within a cell. (C) Population of twelve liver CD45⁺ cell clusters between *Prdm1*^{+/+} and *Prdm1* ^{Δ Ncr1} mice. (D) Dot plot showing the normalized expression of marker genes in liver cNK cells and ILC1s clusters between *Prdm1*^{+/+} and *Prdm1* ^{Δ Ncr1}. The dot size represents the percentage of cells expressing selected genes, and color intensity represents the average expression. (E) Violin plots showing the expression of marker genes *Itga2*, *Itga1*, *Eomes*, and *Tnfrsf10* between liver cNK cells and ILC1s between *Prdm1*^{+/+} and *Prdm1* ^{Δ Ncr1} mice.

Supplemental Figure 6. Cluster-specific markers of liver cNK cell clusters.

(A) Heatmap showing the expression of top 30 upregulated DEGs by log fold change (computed using Wilcox test in the “FindAllMarkers” function of Seurat, avg_log2FC>0.25; *P*<0.05), across the three liver cNK cell clusters (*Prf1*^{hi} cNK, *Junb*^{hi} cNK, *Cxcr3*^{hi} cNK) within *Prdm1*^{+/+} and *Prdm1* ^{Δ Ncr1} mice. (B) Feature plots showing the normalized expression of selected markers for liver cNK cell populations. (C) Violin plots showing the normalized expression of DEGs for each cNK cluster within *Prdm1*^{+/+} and *Prdm1* ^{Δ Ncr1} mice. (D-F) GSEA of the enrichment of Mitochondrial

protein (D), Oxidative phosphorylation (E), and Respiratory electron transport chain (F) in *Junb*^{hi} cNK cluster compared with clusters of *Prf1*^{hi} and *Cxcr3*^{hi} cNK cells. NES, normalized enrichment score.

Supplemental Figure 7. Cluster-specific markers of liver ILC1 clusters.

(A) Heatmap showing the expression of top 30 upregulated DEGs by log fold change (computed using Wilcox test in the “FindAllMarkers” function of Seurat, avg_log2FC>0.25; P<0.05), across the three ILC1 clusters (*Il7r*^{hi} ILC1, *Klra*^{hi} ILC1, *Gzma*^{hi} ILC1) within *Prdm1*^{+/+} and *Prdm1*^{ΔNcr1} mice. (B) Feature plots showing the normalized expression of selected markers for liver ILC1 populations. (C) Normalized expression of DEGs for each ILC1 cluster within *Prdm1*^{+/+} and *Prdm1*^{ΔNcr1} mice. (D) GSEA plots (left) depicting the enrichment of NF-kappa B (D), TNF (E), and MAPK (F) signaling pathways) in *Il7r*^{hi} ILC1 cluster compared with clusters of *Klra*^{hi} and *Gzma*^{hi} ILC1s. Right panel showed dynamic relative expression of the given gene sets from cluster1 to cluster3 between *Prdm1*^{+/+} and *Prdm1*^{ΔNcr1}. Dots represent the average expression of given gene set in each cell, which was calculated through the sum of normalized expression of each individual gene within the designated gene set in every single cell. NES, normalized enrichment score.

Supplemental Figure 8. Identification of four distinct macrophage clusters.

(A) Heatmap showing the top 15 upregulated DEGs across four macrophage clusters (computed using Wilcox test in the “FindAllMarkers” function of Seurat, avg_log2FC>0.25; P<0.05). (B) Percentage of macrophages from *Prdm1*^{+/+} and *Prdm1*^{ΔNcr1} mice among total cells (left) and within clusters (right). (C) Violin plots showing the expression of selected markers of each macrophage cluster within *Prdm1*^{+/+} and *Prdm1*^{ΔNcr1} mice. (D) GO analysis of four distinct macrophage clusters. Dot size represents enriched gene number, and color intensity represents significance.

Supplemental Figure 9. Cell-Cell communication between cNK cells-macrophages and ILC1s-macrophages.

(A and B) Circle plots (A) and cumulative data (B) showing the interaction numbers between cNK cells-macrophages and ILC1s-macrophages. The thickness of the line indicates the number of enrich pairs, and the arrow reflects the direction of the interaction. (C) Heatmap of overall signaling pattern recognized from ligand-receptor pairs between cNK cells-macrophages and ILC1s-macrophages, which contained the sum of signaling from the sender and target cells. (D) Bar graphs showing the information flow in selected active signaling patterns between cNK cells-macrophages and ILC1s-macrophages. Relative information flow was calculated as the sum of the communication probability in given signaling patterns.

Supplemental Figure 10. Ligand-receptor interaction between type I ILCs and macrophages.

Bubble plots showing the significant ligand-receptor pairs between cNK cells, ILC1s, and macrophages within *Prdm1*^{+/+} and *Prdm1* ^{Δ Ncr1} mice. The highlighted Mif-Cd74 and Cxcl-Cxcr signaling was significantly decreased in *Prdm1* ^{Δ Ncr1} mice. Dot size represents the *P*-value, and color intensity represents the communication probabilities. Empty space indicates a communication probability of zero. *P*-value were calculated by the one-sided permutation test.

Supplemental Figure 11: *Prdm1* maintains the IFN- γ production of liver type 1 ILCs in tumor microenvironment.

Representative flow cytometric plot of the frequency of IFN- γ ⁺ liver cNK cells and ILC1s from *Prdm1*^{+/+} and *Prdm1* ^{Δ Ncr1} tumor-bearing mice at day 14 after inoculation with B16F10 melanoma cells via intrasplenic injection (n=5). Data are presented as the mean $\pm$ SEM and were analyzed by 2-tailed, paired t-test. Differences were evaluated between littermates. Each circle and square on graphs represents an individual mouse; *P*, *P*-value; *, *P*<0.05; **, *P*<0.01, ns, not significant.

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Reviewer #1 (Public Review):

He et al. investigate the requirement and function of Blimp1 (encoded by *Prdm1*) in murine NK cells and ILC1. Employing a conditional knockout mouse model (*Prdm1*^{flox} x *Ncr1*^{cre}), the authors describe impaired abundance and maturation of *Prdm1*-deficient NK cells and ILC1 in different tissues. Blimp1-deficient NK cells have reduced expression of cytotoxic molecules (*Gzmb*, *Prf1*) and, in some instances, *Ifng* production, and *Prdm1*^{flox} x *Ncr1*^{cre} mice show impaired tumor control in experimental metastasis models. Using single-cell RNA sequencing analysis, the authors propose that *Prdm1* regulates *JunB* expression and NK cell maturation. Based on *in silico* analyses, the authors suggest manifold intercellular communication between NK/ILC1 and macrophages. Without following up on any of these potentially interesting suggestions, the authors conclude their study reiterating that *Prdm1* regulates IFN γ -production of tumor-infiltrating NK cells and ILC1.

Many of the reported functions of Blimp1 in NK cells have previously been identified using a mixed-chimera strategy comparing *Prdm1* WT and KO NK cells (Kallies et al., Blood 2011). Here, the authors expand on these findings using a conditional model to delete *Prdm1* in NK/ILC1 and single-cell sequencing and provide a more refined analysis of the functions of

Blimp1 in these cells. Cell-chat analysis suggests close interactions of Blimp-dependent NK/ILC1 subsets with hepatic macrophages, but these suggestions are not followed up by experiments. Potentially interesting differences in the macrophage compartment of *Ncr1-Cre* x *Prdm1-fl/fl* mice are suggested by the scRNA-Seq data but are not validated e.g. by FACS. The study falls short in providing new mechanistic insights. Nevertheless, it is an interesting confirmation of "old" suggestions in a more refined setting, and the provided single-cell mRNA-Seq data represents a potentially valuable resource for the community. There are some control analyses that are required to support the conclusions of the authors, and I have a few suggestions that would help to improve the manuscript.

Major comments:

- The authors do not control for the potential effects of Cre expression. Expression of Cre from within the *Ncr1* locus (using the mouse model established by Narni-Mancinelli et al.) has significant effects on NK cells and especially ILC1s (reducing their frequency and absolute numbers and altering their functionality). The authors should characterize the *Ncr1cre* mice used here (developed by Shanghai Model Organism Center) in this regard and should use proper controls (*Ncr1Cre⁺ Prdm1wt/wt* as control for *Ncr1Cre⁺ Prdm1fl/fl*, instead of WT littermates) for all of their key data, e.g. those depicted in Fig 1FG, 2ADFH, 7D, S2,3,4.
- Several of the phenotypic findings on NK cells have been described before by Kallies et al. in 2011 (Ref 29), although using a different genetic *Prdm1*-ablation model (*Prdm1-GFP/GFP* knockin/knockout model). This study reported impaired NK cell maturation, reduced *Gzmb* expression, impaired *in vivo* cytotoxicity against subcutaneous RMA-S cells, impaired *in vitro* proliferation, comparable *in vitro* killing, increase in BM NK cell numbers. The authors should discuss/mention this more prominently in their manuscript, and highlight where they confirm or refine these previous findings, and where they actually provide new information.
- What is the reason to refer to the enriched cluster in *Blimp1*-deficient NK cells as "Junbhi"? There is no follow-up for a function of *Junb*, and there are many other genes upregulated in these cells. Most critically, these cells seem to represent exactly the c-Kithi cells that Kallies et al. already showed and discussed in their paper. The authors should stain for *Kit*, and also refer to this. Also, MacKay et al. performed *Blimp1*-Chip-Seq (in T cells), maybe it would be interesting to check to which of the identified DEGs *Blimp1* can bind.
- cNK cells are considered circulating cells, that transiently pass through the liver. Previous studies have suggested almost identical gene expression patterns in hepatic and splenic NK cells. In functional tests, they often "perform" identically. I am therefore a bit surprised that the authors find a differential dependency of *Blimp1* for the IFN γ production of splenic (no role of *Blimp1*) versus hepatic (*Blimp1* regulating IFN γ production) NK cells (Fig S3). Do the authors have any suggestions on that? The analyses are performed by 12+4h stimulations with IL12/18, which could involve the effects of altered bystander cells (as suggested by Figure 6). Therefore, these analyses should be provided upon standard 4h stimulations with IL12/18 and also with PMA/I under BFA. Note: liver and splenic cNK cells look quite different in the chosen histograms in Figures 7 A, B, C, yet there is massive variability in these analyses
- is there any systematic/technical problem?
- Figure 4 H/I - In contrast to NK cells in Fig 4E, F, the KO and WT ILC1s seem to co-cluster largely. Authors should validate differentially expressed genes. How strong is the effect of *Blimp1* in ILC1s? Or is *Blimp1* a critical TF driving effector differentiation in NK cells, while it has only subtle effects in ILC1 (these may be regulated by *Hobit*)? This seems an interesting finding that should at least be discussed. For these types of small differences in ILC1, FACS confirmation analyses should be performed and findings be reevaluated using Cre-expressing controls (see above).

The authors describe and discuss some of Figure 1 and 2 data as if Blimp1 would be involved in alternative NK versus ILC1 fates, but there is no evidence for this.

- There are several recent studies suggesting a role for Hobit, homologue of Blimp1, in NK cells and in ILC1, and in the control of liver metastases. The authors should discuss similar and unique functions of Hobit and Blimp1, also in the regulation of gene expression patterns, and should refer to these studies.

- Figure 4: The authors should discuss (and cross-validate) their liver gene expression analyses in the context of published datasets of NK and ILC1, such as the ones by Lopez et al, Friedrich et al, Ducimetiere et al and Yomogida et al.

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Reviewer #2 (Public Review):

Summary:

This study offers a significant advancement in understanding liver innate lymphoid cell (ILC) biology by elucidating the role of the transcription factor Prdm1. It shows that Prdm1 is crucial in maintaining the balance between conventional natural killer (cNK) cells and ILC1s in the liver, with knockout models revealing a vital role in cancer defense mechanisms. Despite not affecting direct cytotoxicity, Prdm1 deficiency leads to increased cancer metastasis and reduced secretion of key molecules like IFN- γ , pointing to its importance in immune regulation. The use of single-cell RNA sequencing further underscores Prdm1's role in cellular communication within the liver's immune milieu. This study is a robust contribution to the field, providing insights that could inform new immunotherapy approaches for liver cancer.

Strengths:

The study's strength lies in its comprehensive approach, combining the specificity of Prdm1 conditional deletion in Ncr1-cre mice with integrative omics analyses and cutting-edge cytometry to delineate Prdm1's role in liver Type 1 ILC biology and its functional implications in tumor immunity. This multifaceted strategy not only clarifies Prdm1's influence on ILC composition and maturation but also conveys potential therapeutic insights for liver cancer immunotherapy.

Weaknesses:

A notable weakness of the study is the limited scope of in vivo disease models, primarily relying on the B16F10 melanoma model, which may not fully capture the complex behavior of Type 1 ILCs across diverse cancer types. Furthermore, the absence of direct human data, such as the effects of PRDM1 deletion in human NK cells or stem cells during their differentiation into NK and ILC1, leaves a gap in translating these findings to clinical settings.

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